

Lili Zheng

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WORK EXPERIENCE

Assistant Professor

07/2024 – Now

Department of Statistics, University of Illinois Urbana-Champaign

Postdoctoral researcher

06/2021 – 05/2024

Department of Electrical and Computer Engineering, Rice University

Advisor: Genevera I. Allen

EDUCATION

University of Wisconsin - Madison

2016 – 2021

Ph.D., Statistics

Advisor: Garvesh Raskutti

University of Science and Technology of China

2012 – 2016

B.S., Statistics

RESEARCH INTEREST

Interpretable machine learning, distribution-free inference, multiple testing, data integration, ensemble methods, spectral methods, missing data, tensor data analysis, graphical models, time series

HONORS

Gold Medal Young Researcher Poster Competition Award,

Conference on Recent Advances in Statistics and Data Science at Rutgers.

2023

IMS Hannan Graduate Student Travel Award.

2021

Travel grant from Institute for Foundations of Data Science (IFDS), UW-Madison.

2019

Travel grant from IMA workshop on Forecasting from Complexity.

2018

Honorable Mention in Mathematical Contest of Modeling, COMAP (Top 20%)

2015

National scholarship, USTC. (Top 2%)

2015

Outstanding freshman scholarship, USTC(Top 20%)

2012

PUBLICATIONS

(*: equal contribution; †: student advisee.)

Preprints

18. Y. Cheng[†], **L. Zheng**, “LOCO-AdaMP: Built-in LOCO Inference for Adaptive Minipatch Ensembles with Enhanced Prediction.” *Under review*, <https://arxiv.org/abs/2609.36396>.
17. X. Liu[†], **L. Zheng**, “Model-Agnostic Feature Selection via LOCO-Guided Adaptive Minipatch Sampling.” *Under review*, <https://arxiv.org/abs/2609.24126>.
16. J. Liu*, **L. Zheng***, Z. Zhang, G. I. Allen, “Joint Semi-Symmetric Tensor PCA for Integrating Multi-modal Populations of Networks”. *Under revision*, <https://arxiv.org/abs/2312.14416>.
15. **L. Zheng**, G. Raskutti, “High-dimensional Multi-class Classification with Presence-only Data”. <https://arxiv.org/abs/2304.09305>
14. L. Gan*, **L. Zheng***, G. I. Allen, LOCO Feature Importance Inference without Data Splitting via Minipatch Ensembles”. *Under revision*, <https://arxiv.org/abs/2206.02088>.

13. A. Chang, **L. Zheng**, G. I. Allen, “Low-Rank Covariance Completion for Graph Quilting with Applications to Functional Connectivity”. *Under review*, <https://arxiv.org/abs/2209.08273>.

Peer-reviewed Journal Publications

11. C. O. Little, **L. Zheng**, G. I. Allen, “iLOCO: Distribution-Free Inference for Feature Interactions”. *Accepted to Transactions on Machine Learning Research*, <https://arxiv.org/abs/2502.06661>.
10. **L. Zheng**^{*}, A. Chang^{*}, G. I. Allen, “Cluster Quilting: Spectral Clustering for Patchwork Learning”. *Accepted to the Journal of the American Statistical Association, Theory and Methods*, <https://arxiv.org/pdf/2406.13833>.
9. **L. Zheng**, G. I. Allen, “Graphical Model Inference with Erosely Measured Data”, *Journal of the American Statistical Association, Theory and Methods*, 2023.
8. A. Chang^{*}, **L. Zheng**^{*}, G. Dasarathy, G. I. Allen, “Nonparanormal Graph Quilting with Applications to Calcium Imaging”, *STAT*, 2023.
7. G. I. Allen, L. Gan, **L. Zheng**, “Interpretable Machine Learning for Discovery: Statistical Challenges & Opportunities”, *Annual Review of Statistics and Its Application*, 2023.
6. H. Chen^{*}, **L. Zheng**^{*}, R. A. Kontar, G. Raskutti (*: equal contribution), “Gaussian Process Parameter Estimation Using Mini-batch Stochastic Gradient Descent: Convergence Guarantees and Empirical Benefits”, *Journal of Machine Learning Research*, 2022.
5. Y. Zhou, A. R. Zhang, **L. Zheng**, Y. Wang, “Optimal High-order Tensor SVD via Tensor-train Orthogonal Iteration”, *IEEE Transactions on Information Theory*, 2022.
4. **L. Zheng**, G. Raskutti, R. Willett, B. Mark, “Context-dependent Networks in Multivariate Time Series: Models, Methods, and Risk Bounds in High Dimensions”, *Journal of Machine Learning Research*, 2021.
3. **L. Zheng**, G. Raskutti, “Testing for High-dimensional Network Parameters in Auto-regressive Models”, *Electronic Journal of Statistics*, 2019.

Peer-reviewed Conference Publications

3. **L. Zheng**, Z. T. Rewolinski, G. I. Allen, “A Low-Rank Tensor Completion Approach for Imputing Functional Neuronal Data from Multiple Recordings”, *IEEE Data Science and Learning Workshop (DSLW)*, 2022.
2. **L. Zheng**, G. I. Allen, “Learning Gaussian Graphical Models with Differing Pairwise Sample Sizes”, *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022.
1. H. Chen^{*}, **L. Zheng**^{*}, R. A. Kontar, G. Raskutti (*: equal contribution), “Stochastic Gradient Descent in Correlated Settings: A Study on Gaussian Processes”, *Neural Information Processing Systems (NeurIPS)*, 2020.

SOFTWARE

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- “tt_TTOI: Tensor-Train Orthogonal Iteration”, contribute python functions to the **Python** library **TensorLy**, 2022.
 - “Network inference for autoregressive models”, **R** and **Matlab** functions with tutorials, 2020.

PROFESSIONAL SERVICE

- Organizer for an invited session at the Joint Statistical Meetings, 2025
- Session chair for International Indian Statistical Association (IISA) Conference, 2025
- Session chair for IMSI workshop: Statistics Meets Tensors, 2025
- Organizer for a topic-contributed session in Joint Statistical Meetings, 2022
- Session chair for ICASSP, 2022
- Journal referee (48 in total, # in parenthesis): *Annals of Applied Statistics* (4), *Biometrika* (3), *Computational Statistics and Data Analysis* (2), *Electronic Journal of Statistics* (1), *IEEE Transactions on Information Theory* (3), *Journal of the American Statistical Association* (4), *Journal of Computational and Graphical Statistics* (3), *Journal of Machine Learning Research* (13), *Journal of Multivariate Analysis* (5), *Journal of the Royal Statistical Society: Series B* (6), *SIAM Journal on Mathematics of Data Science* (1), *Scandinavian Journal of Statistics* (1), *Statistical Sinica* (1), *ACM Transactions on Probabilistic Machine Learning* (1).
- Conference reviewer: *International Conference on Artificial Intelligence and Statistics (AISTATS) 2021*, *International Conference on Machine Learning (ICML) 2022*.

TEACHING AND MENTORING EXPERIENCE

Teaching

Stat 425 (Statistical Modeling I), Instructor	Spring 2026
Stat 425 (Statistical Modeling I), Instructor	Spring 2025
The Summer Institute for Statistics in Big Data, Teaching Assistant	Summer 2023
Stat 301 (Introduction to Statistical Methods), Teaching Assistant	Fall 2016/ Spring 2017

Mentoring

1. Xuhui Liu, PhD candidate, Statistics, UIUC.
2. Yinan Cheng, UIUC, PhD candidate, Statistics, UIUC.
3. Camille Little, PhD candidate, Electrical and Computer Engineering, Rice University.
4. Jiaming Liu, PhD candidate, Statistics, Rice University.
5. Quan Le, former undergraduate student, Computer Science and Mathematics, Rice University, 2023; now a PhD student at Yale.
6. Zach Rewolinski, former undergraduate student, Statistics and Computer Science, Rice University, 2023; now a PhD student at UC Berkeley.

PRESENTATIONS

- Invited talk at the 4th Joint Conference on Statistics and Data Science (JCSDS 2026)
- Poster presentation at 2026 AISTATS (journal-to-conference track), 05/03/2026
- Contributed talk at 2025 IMS International Conference on Statistics and Data Science (ICSIDS), 12/16/2025
- Invited talk at CFE-CMStatistics, 12/15/2025
- Invited talk at EcoSta, 08/22/2025
- Invited talk at the IISA conference, 06/12/2025

- Poster presentation at IMSI workshop: Statistics Meets Tensors, 05/06/2025
- Invited talk at 2024 IMS International Conference on Statistics and Data Science (ICSIDS), 12/16/2024
- Invited talk at CFE-CMStatistics, 12/15/2024
- Invited talk at Cornell University, 2/28/2024
- Invited talk at the University of California, San Diego, 2/15/2024
- Invited talk at the University of British Columbia, 2/8/2024
- Invited talk at the University of Waterloo, 2/5/2024
- Invited talk at the University of California, Irvine, 1/30/2024
- Invited talk at the University of Illinois at Urbana-Champaign, 1/25/2024
- Invited talk at the University of Michigan, 1/19/2024
- Invited talk at the University of California, San Diego, 1/12/2024
- Invited talk at the University of Florida, 12/6/2023
- Contributed talk at Joint Statistical Meetings, 2023
- Poster presentation at “Conference on Recent Advances in Statistics and Data Science,” Rutgers University, 2023
- Poster presentation at the conference “Statistical Foundations of Data Science and their Applications,” Princeton University, 2023
- Talk in a topic-contributed session at Joint Statistical Meetings, 2022
- Poster presentation at “Workshop on Distribution-Free Uncertainty Quantification,” ICML, 2022
- Talk at IEEE Data Science and Learning Workshop, 2022
- Poster presentation at ICASSP, 2022
- Poster presentation at the “Conference on Advances in Bayesian and Frequentist Statistics,” Rutgers University, 2022
- Poster presentation at the workshop “Perspectives in Statistical Modeling and Inference,” University of Pennsylvania, 2021
- Poster presentation at Joint Statistical Meetings, 2020
- Poster presentation at NeurIPS, 2020
- Poster presentation at Joint Statistical Meetings, 2019